AI & GENAI HANDS-ON PROJECT

AI-BASED ELECTRICAL FAULT DETECTION SYSTEM

Submitted By:

1. Student Name- Jigyasha Kumari]
2. Institution Name- Government polytechnic jajpur
3. Department- Electrical engg.
4. Registration no. - F25126002022

Edit with WPS Office

edunet foundation

OUTLINE

- **Problem Statement** (the real-world/classroom problem you addressed)
- **Proposed AI/GenAI Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result / Demo Output**
- **Conclusion**
- **Future Scope**
- **References**

Edit with WPS Office

PROBLEM STATEMENT

- **Electrical faults are hard to catch early.** Overloads, short circuits and abnormal currents damage equipment, cause downtime and create fire and shock hazards.
- **Fixed-threshold protection is limited:** fuses and simple relays react only after a limit is crossed, cannot tell fault types apart, and give no warning of developing problems.
- **Manual inspection does not scale:** it is periodic, slow and depends on technician skill, so faults inside homes, labs and small industries often go unnoticed.
- **Goal:** use AI on live sensor data to detect a fault the moment it appears and classify its type (overload, short circuit, abnormal current) so the right action can be taken.

Edit with WPS Office

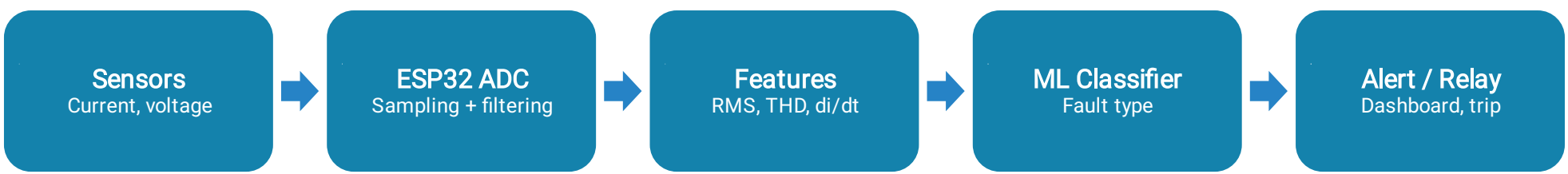
PROPOSED AI/GENAI SOLUTION

- **Track A — Machine Learning:** a supervised classifier trained on electrical sensor data.
- **Model type and classes**
  - Random Forest / XGBoost classifier on engineered features; optional 1D-CNN or LSTM on raw current windows for higher accuracy.
  - Classes: Normal, Overload, Short-circuit, Abnormal current (harmonics, imbalance, sag/surge).
- **Data collection and training approach**
  - Current and voltage readings from sensors (CT / ACS712, ZMPT101B), plus simulated fault data from MATLAB/Simulink or a public fault dataset.
  - Windowed readings are converted to features (RMS, peak, crest factor, THD, di/dt, phase imbalance), labelled, split into train/test sets and used to train the model.
- **Outcome:** a system that flags a fault in real time, names its type and raises an alert or trips a relay.

Edit with WPS Office

SYSTEM APPROACH

- **Platform / tool used:** Python (scikit-learn, XGBoost, TensorFlow Lite) for training; ESP32 / Raspberry Pi for data acquisition and inference.
- **Libraries, APIs and datasets:** NumPy, Pandas, SciPy (FFT), Matplotlib; MQTT for data transfer; Grafana or a web dashboard; MATLAB/Simulink-simulated faults and public electrical fault datasets.
- **Hardware:** current sensor (CT / ACS712), voltage sensor (ZMPT101B), relay module, buzzer / indicator LEDs.



Edit with WPS Office

APPROACH & DEPLOYMENT

- **Tool / Model Selection:**
  - Random Forest / XGBoost chosen for fast, accurate and explainable classification on tabular features; a TFLite version runs on the ESP32 / Raspberry Pi.
- **Input / Data:**
  - Sampled current and voltage windows, converted to RMS, peak, crest factor, THD, di/dt and phase imbalance, labelled as Normal, Overload, Short-circuit or Abnormal current.
- **Build / Training Process:**
  - Data cleaned and scaled, split into train/test sets, hyperparameters tuned with cross-validation, and results checked with a confusion matrix.
- **Output / Prediction Process:**
  - Live sensor windows are scored in real time; the predicted class is shown on the dashboard, and a fault triggers an alert and the relay output.
- **Note:** hardware protection (fuse / breaker) stays in place; the AI layer detects, classifies and alerts.

Edit with WPS Office

RESULT

- The system reads live sensor data and labels each window with a fault class:

Normal
Steady RMS current and voltage, low THD, balanced phases

Overload
Sustained rise in RMS current above rated load

Short-circuit
Sudden current spike with very high di/dt and voltage dip

Abnormal current
High harmonics (THD), imbalance or sag / surge

Dashboard shows live current, voltage, predicted class and an alert when a fault is detected.

[Add your measured accuracy, confusion matrix and demo screenshot here]

Edit with WPS Office

FUTURE SCOPE

- **Deep learning models:** 1D-CNN / LSTM on raw waveforms for higher accuracy and earlier detection.
- **Predictive maintenance:** forecast developing faults and equipment ageing before failure occurs.
- **Fault location and severity:** identify which circuit or phase is affected and how serious the fault is.
- **IoT and cloud integration:** multi-site monitoring, SMS / app alerts and remote relay control.
- **Scale-up:** three-phase and industrial systems, solar / microgrid protection, and edge deployment on low-cost microcontrollers.

Edit with WPS Office

REFERENCES

- scikit-learn documentation: scikit-learn.org
- XGBoost documentation: xgboost.readthedocs.io
- TensorFlow Lite for Microcontrollers: tensorflow.org/lite/microcontrollers
- Kaggle: Electrical Fault Detection and Classification dataset
- MATLAB / Simulink Power Systems fault simulation: mathworks.com
- ESP32 documentation: docs.espressif.com

Edit with WPS Office

CONCLUSION

- **What was built:** an AI-based system that reads current and voltage sensor data, detects electrical faults and classifies them as overload, short circuit or abnormal current.
- **Why it works:** engineered features such as RMS, THD and di/dt separate fault types better than a single fixed threshold.
- **Challenges:** limited real fault data, sensor noise and safe fault testing. Simulated data and filtering helped; more real-world data would improve robustness.
- **Application:** the same approach applies in labs, classrooms and small industries as a low-cost monitoring layer alongside standard protection devices.

Edit with WPS Office

CREDLY CREDENTIAL / PROJECT LINK

- Credly Badge Link: [paste your Credly credential URL]
- Project/Demo Link (optional): https://claude.ai/artifact/BnC1wkx2dtaahNeaHor5Bv

Edit with WPS Office

THANK YOU

Edit with WPS Office